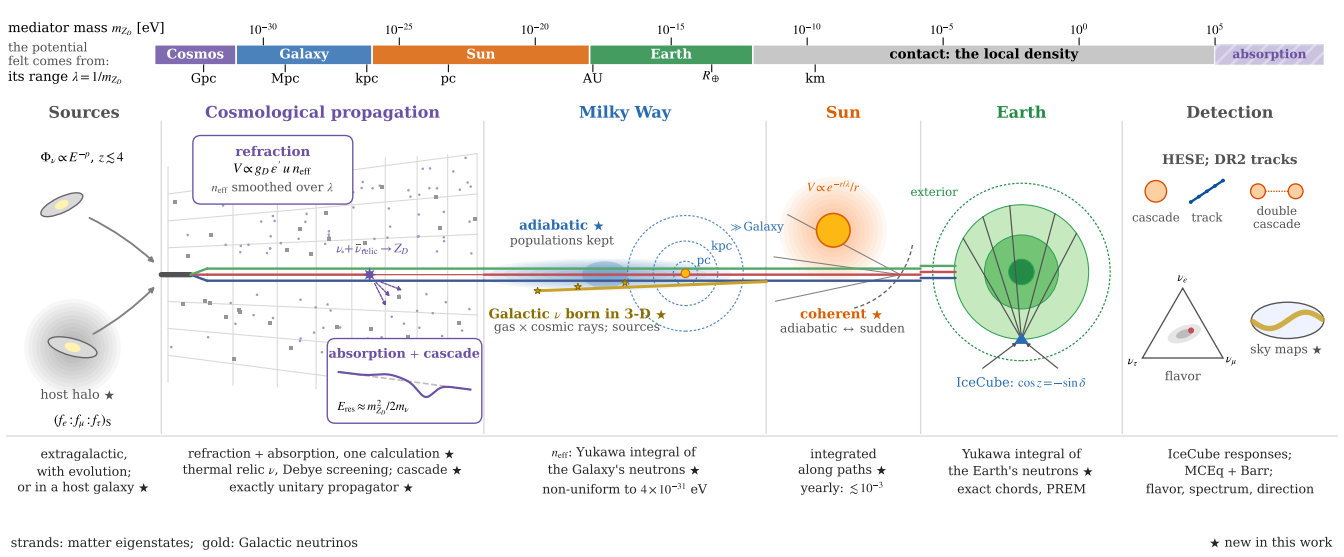

mediator mass m_{Z_0} [eV]
10^{-30}
10^{-25}
10^{-20}
10^{-15}
10^{-10}
10^{-5}
10^{0}
10^{5}
the potential felt comes from:
its range $\lambda = 1/m_{Z_0}$
Cosmos
Galaxy
Sun
Earth
contact: the local density
absorption
Gpc
Mpc
kpc
pc
AU
$R_\oplus$
km
Sources
Cosmological propagation
Milky Way
Sun
Earth
Detection
$\Phi_\nu \propto E^{-\eta}$, $z \lesssim 4$
refraction
$V \propto g_D\, e^{\cdot}\, u\, n_{\rm eff}$
$n_{\rm eff}$ smoothed over λ
$\nu_\tau + \bar{\nu}_{\rm relic} \to Z_D$
absorption + cascade
$E_{\rm res} \approx m_{Z_0}^2/2m_\nu$
host halo ★
$(f_e : f_\mu : f_\tau)$s
adiabatic ★
populations kept
≫ Galaxy
kpc
pc
Galactic ν born in 3-D ★
gas × cosmic rays; sources
$V \propto e^{-r/\lambda}/r$
coherent ★
adiabatic ↔ sudden
exterior
IceCube: $\cos z = -\sin\delta$
HESE; DR2 tracks
cascade
track
double cascade
ν_e
ν_τ
ν_μ
flavor
sky maps ★
extragalactic, with evolution; or in a host galaxy ★
refraction + absorption, one calculation ★
thermal relic ν, Debye screening; cascade ★
exactly unitary propagator ★
$n_{\rm eff}$: Yukawa integral of the Galaxy's neutrons ★
non-uniform to 4×10^{-31} eV
integrated along paths ★
yearly: $\lesssim 10^{-3}$
Yukawa integral of the Earth's neutrons ★
exact chords, PREM
IceCube responses; MCEq + Barr;
flavor, spectrum, direction
strands: matter eigenstates; gold: Galactic neutrinos
★ new in this work